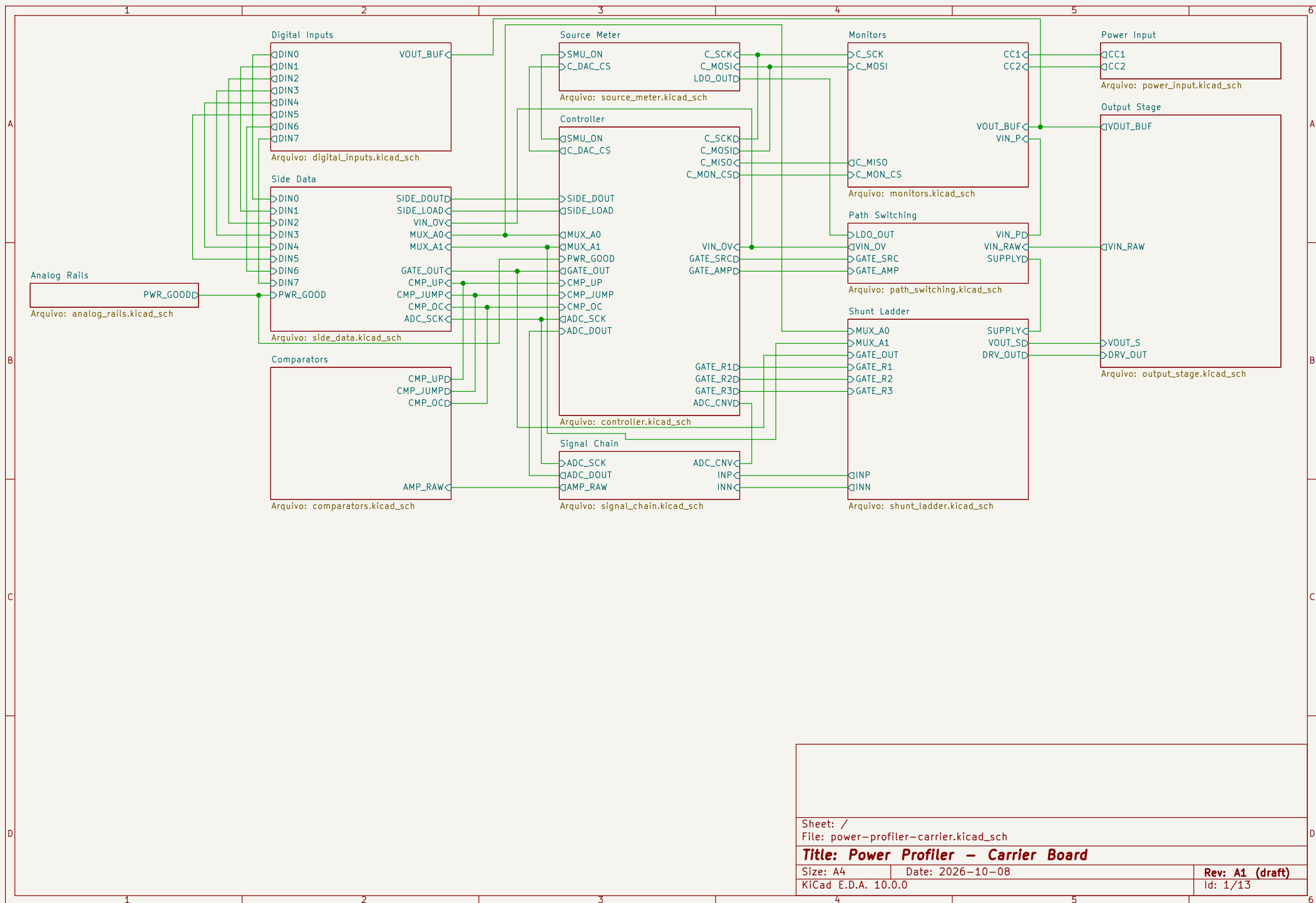
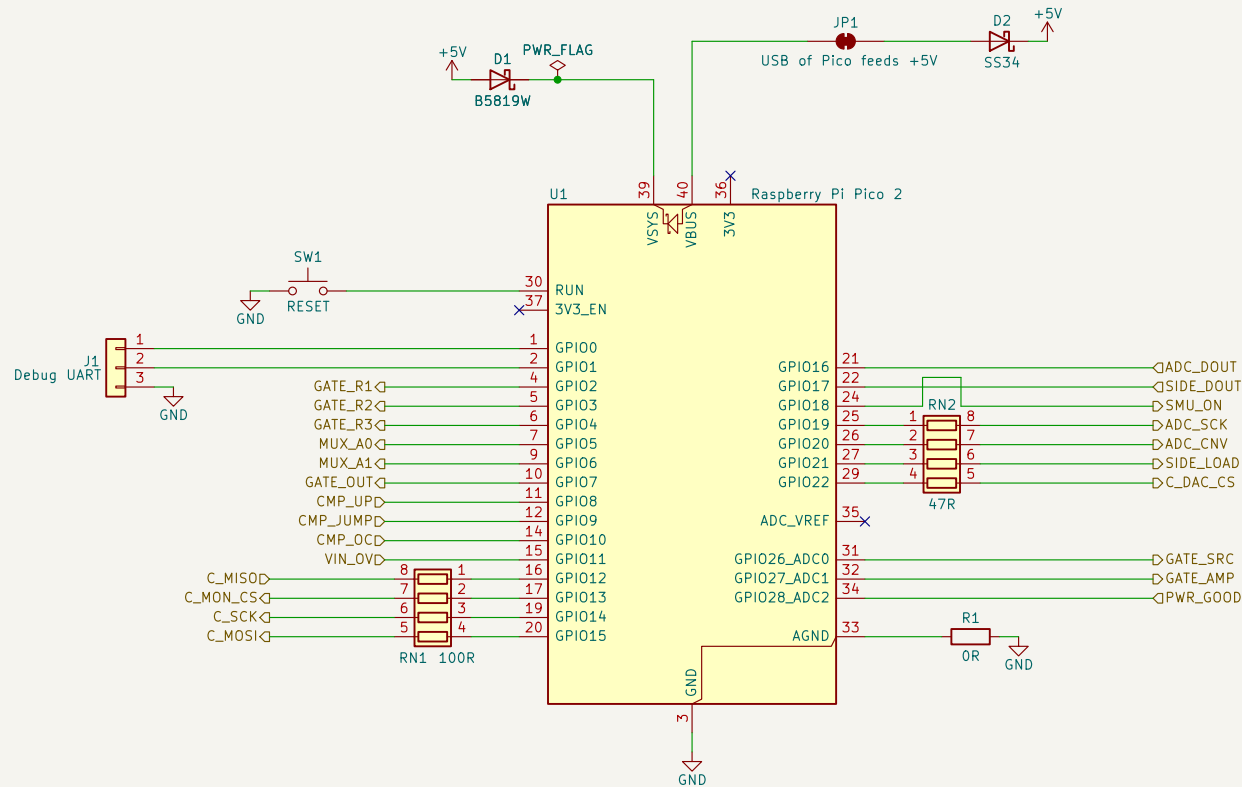




U1 is a Raspberry Pi Pico 2 on two 1x20 sockets, the only programmable part of the instrument. A Pico 2 W fits the same sockets. PIO block 0 (GP16, GP17, GP19 to GP21) clocks the converter and reads its data and the side data in the same word. PIO block 1 (GP2 to GP10) is the range sequencer: comparators in, gates and multiplexer address out. Reset or no firmware: every pin is an input pulled low, and the pull-downs at the gate drivers and at the regulator enable keep every switch open.



Power: the USB connector of the Pico feeds +5V through JP1 and D2, and +5V feeds VSYS through D1, so one cable is enough for small loads. Cut JP1 to keep the carrier off the USB port of the computer.

Raspberry Pi Pico 2 (RP2350) on pin sockets: the only programmable part

Sheet: /Controller/
File: controller.kicad_sch

Title: Controller: Raspberry Pi Pico 2

Size: A4

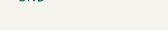
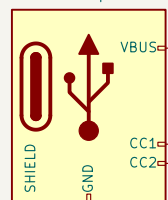
Date: 2026-10-08

Rev: A1 (draft)

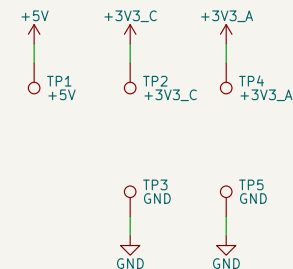
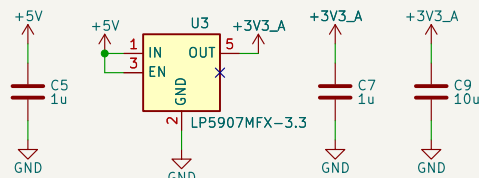
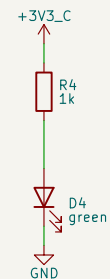
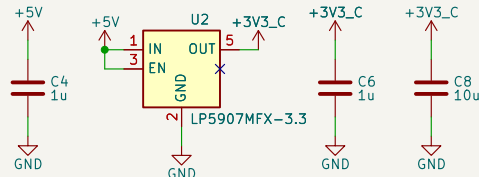
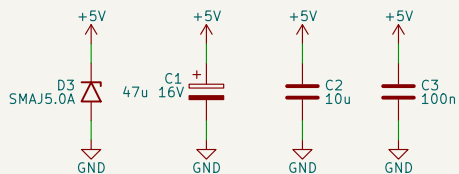
KiCad E.D.A. 10.0.0

Id: 2/13

J2
USB-C power



Source capability from the CC voltage (firmware, through the monitor converter):
0.25 V to 0.61 V default USB, 0.70 V to 1.16 V 1.5 A, 1.31 V to 2.04 V 3 A.
+3V3_C: logic of the carrier. +3V3_A: converter, its driver and the comparators.
H1 to H4: mounting holes, tied to ground.



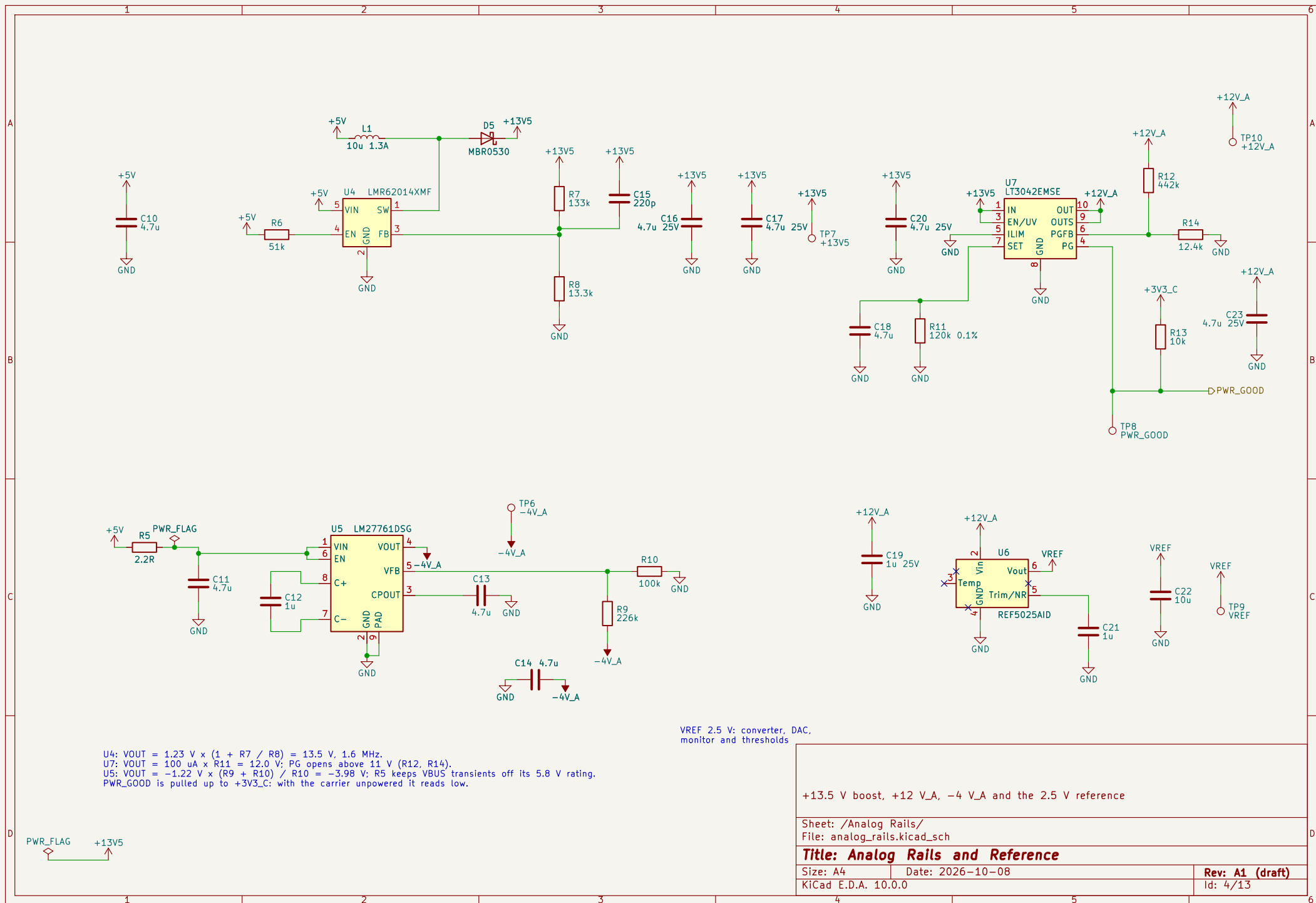
USB-C sink, 5 V only; CC voltages go to the monitor converter

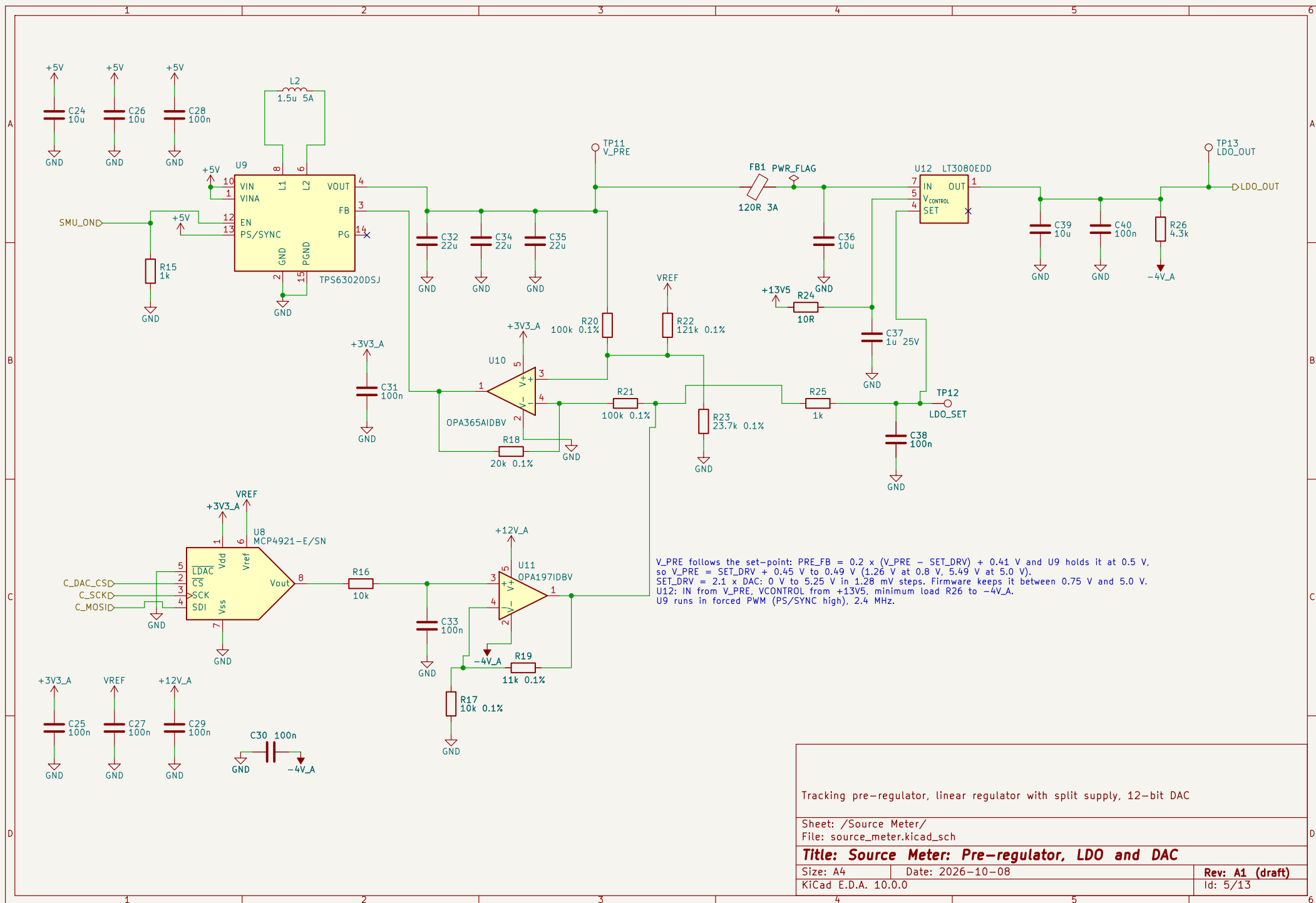
Sheet: /Power Input/
File: power_input.kicad_sch

Title: Power Input and Logic Supplies

Size: A4 Date: 2026-10-08
KiCad E.D.A. 10.0.0

Rev: A1 (draft)
Id: 3/13





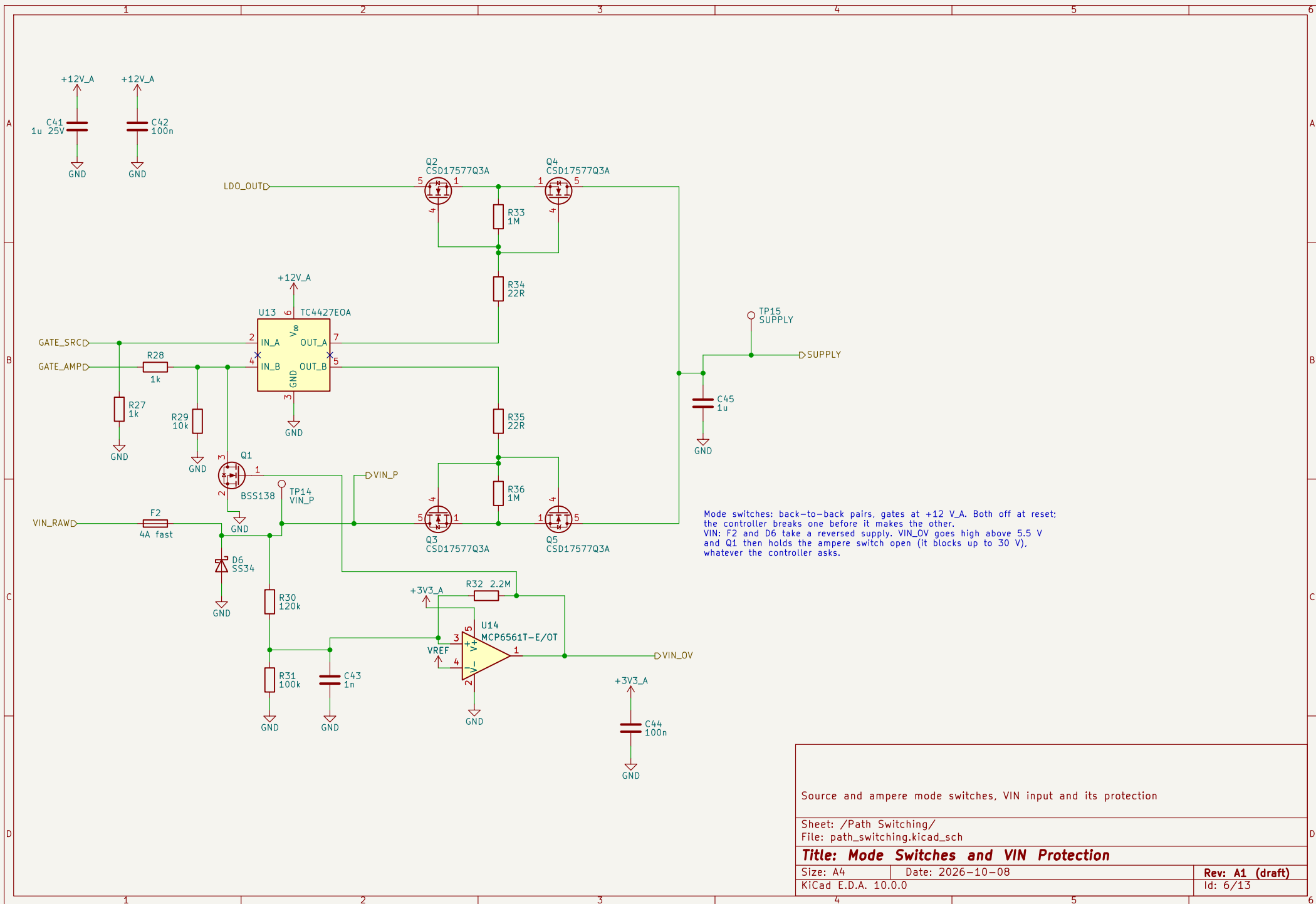
Tracking pre-regulator, linear regulator with split supply, 12-bit DAC

Sheet: /Source Meter/
 File: source_meter.kicad_sch

Title: Source Meter: Pre-regulator, LDO and DAC

Size: A4
 Date: 2026-10-08
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 Id: 5/13



Source and ampere mode switches, VIN input and its protection

Sheet: /Path Switching/
File: path_switching.kicad_sch

Title: Mode Switches and VIN Protection

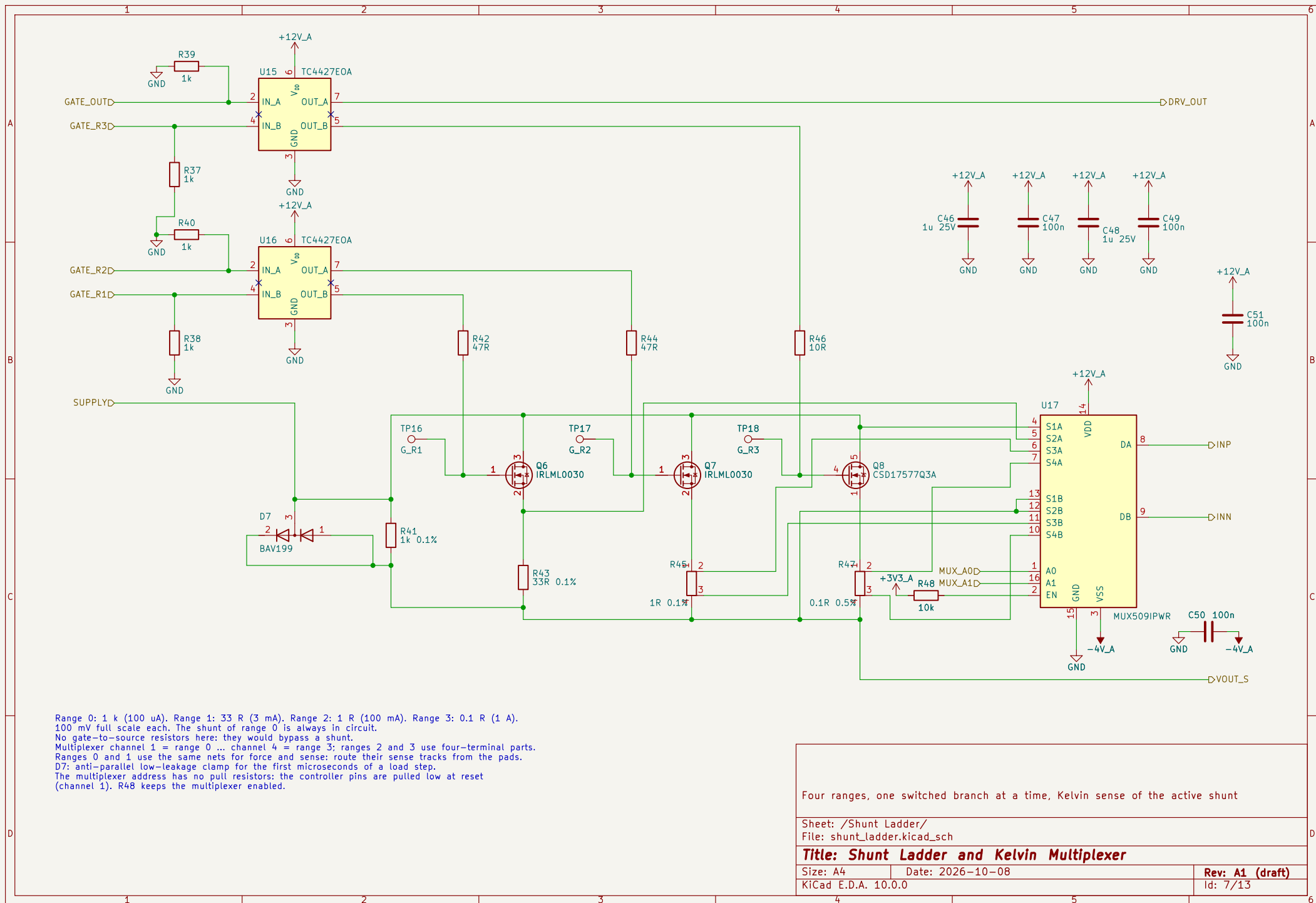
Size: A4

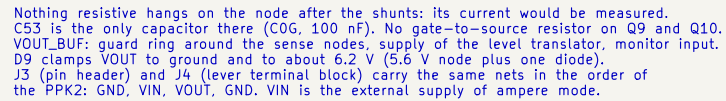
Date: 2026-10-08

Rev: A1 (draft)

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Id: 6/13





Sheet: /Output Stage/
File: output_stage.kicad_sch

Title: Output Switch, DUT Connectors and Guard Buffer

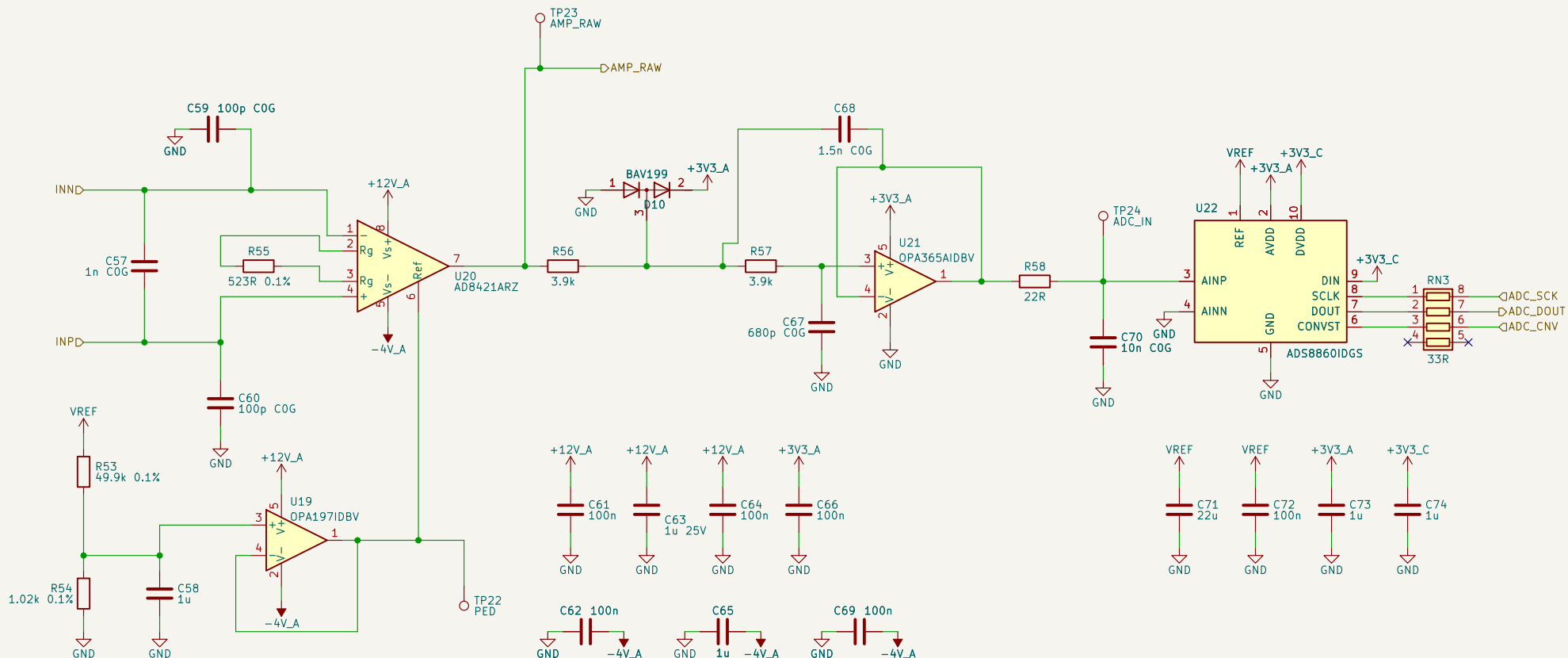
Size: A4

Date: 2026-10-08

Rev: A1 (draft)

Size: A1	1
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Id: 8/13



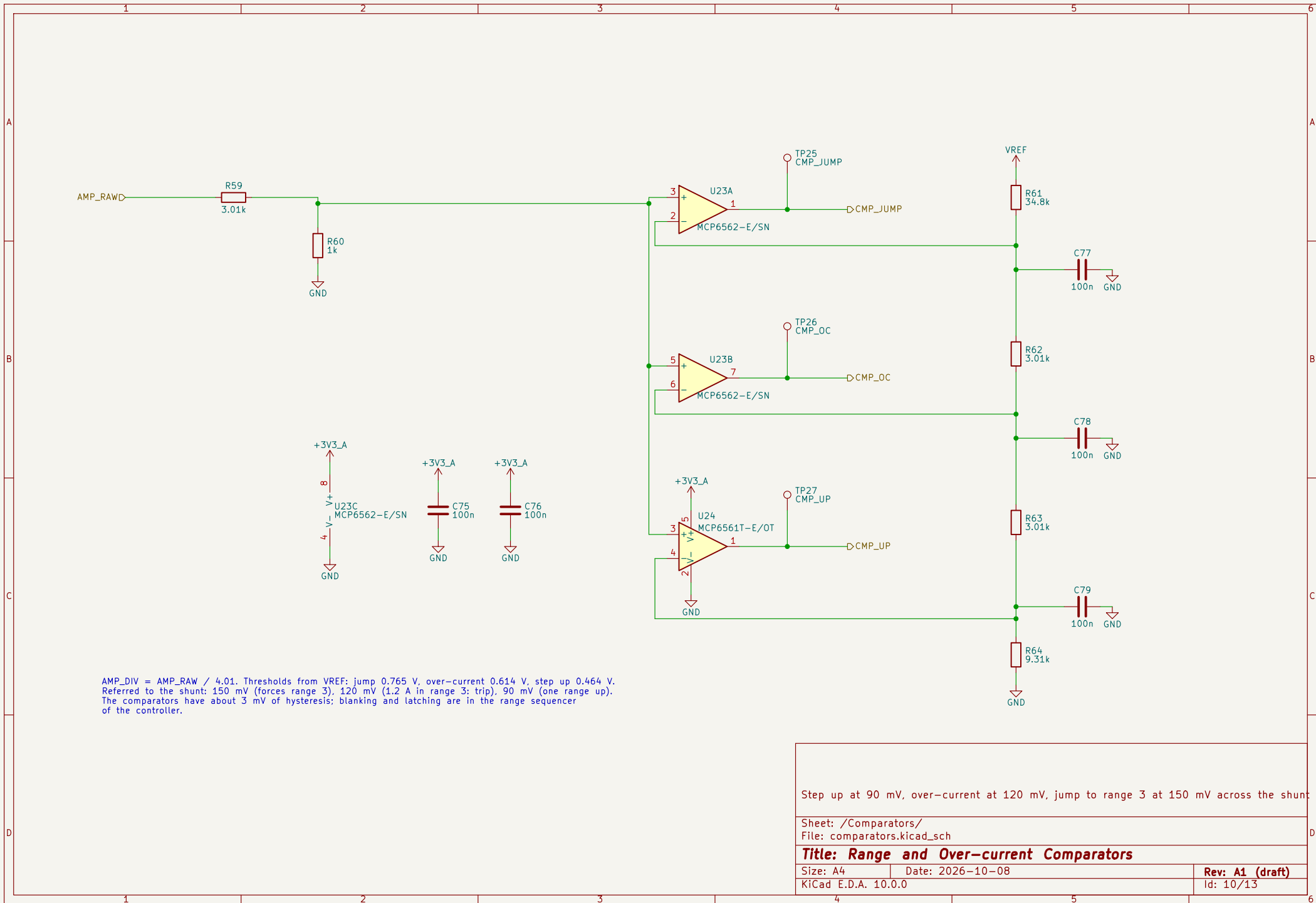
U20: $G = 1 + 9.9 \text{ k} / R55 = 19.93$. Reference pin at 50 mV: zero current reads about 1300 codes.
 AMP_RAW can reach 10 V during a range change; R56 and D10 limit the driver input.
 U21: Sallen-Key, 40 kHz, $Q = 0.74$. R58 and C70 as the converter input needs.
 U22: 3-wire mode, DIN high, CONVST is the I2S word clock, SCLK the bit clock.
 C71 and C72 sit at the REF pin.

Gain 19.93, 50 mV pedestal, limiter, 40 kHz two-pole filter, 16-bit converter

Sheet: /Signal Chain/
 File: signal_chain.kicad_sch

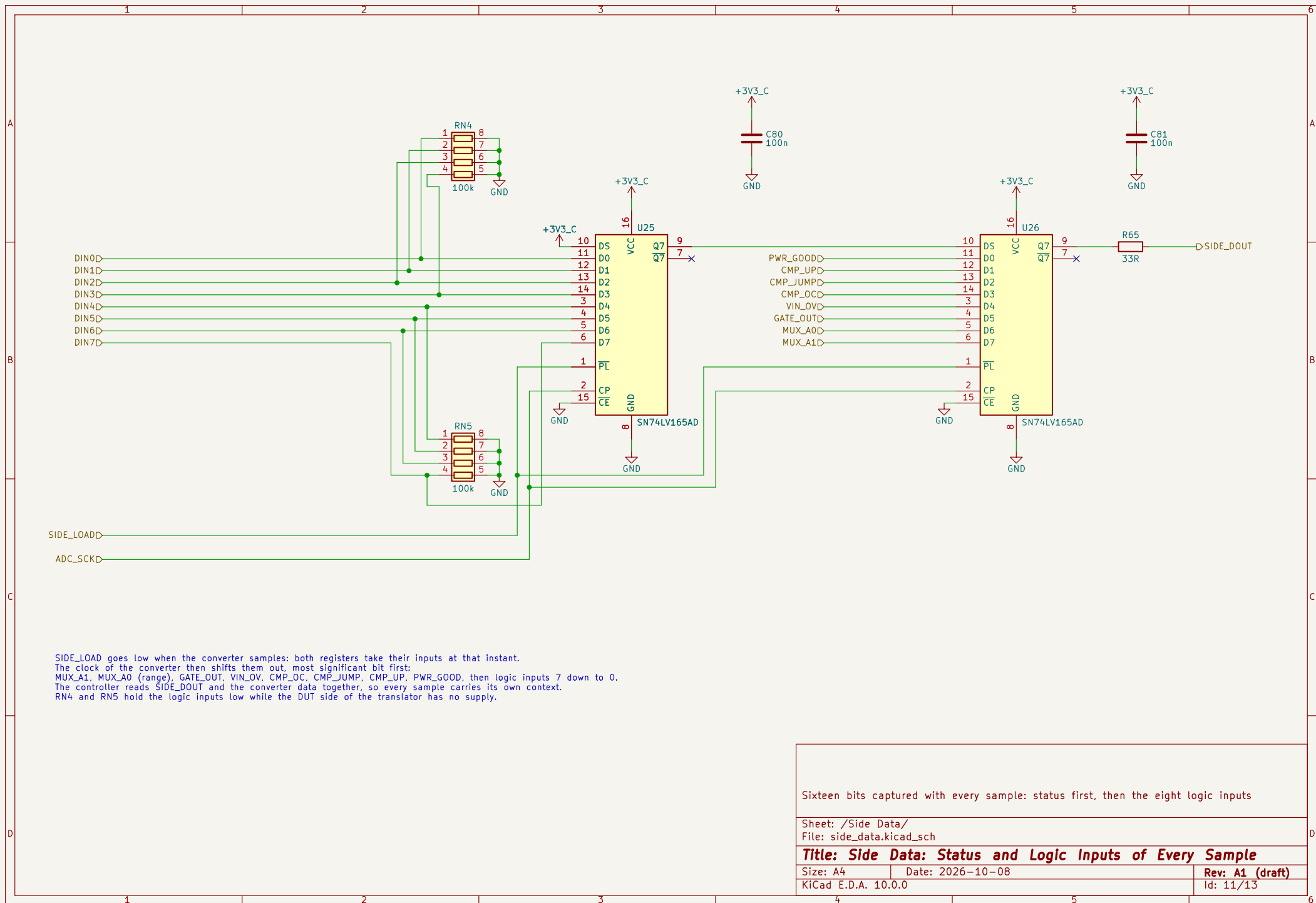
Title: Amplifier, Anti-alias Filter and ADC

Size: A4	Date: 2026-10-08	Rev: A1 (draft)
KiCad E.D.A. 10.0.0		Id: 9/13

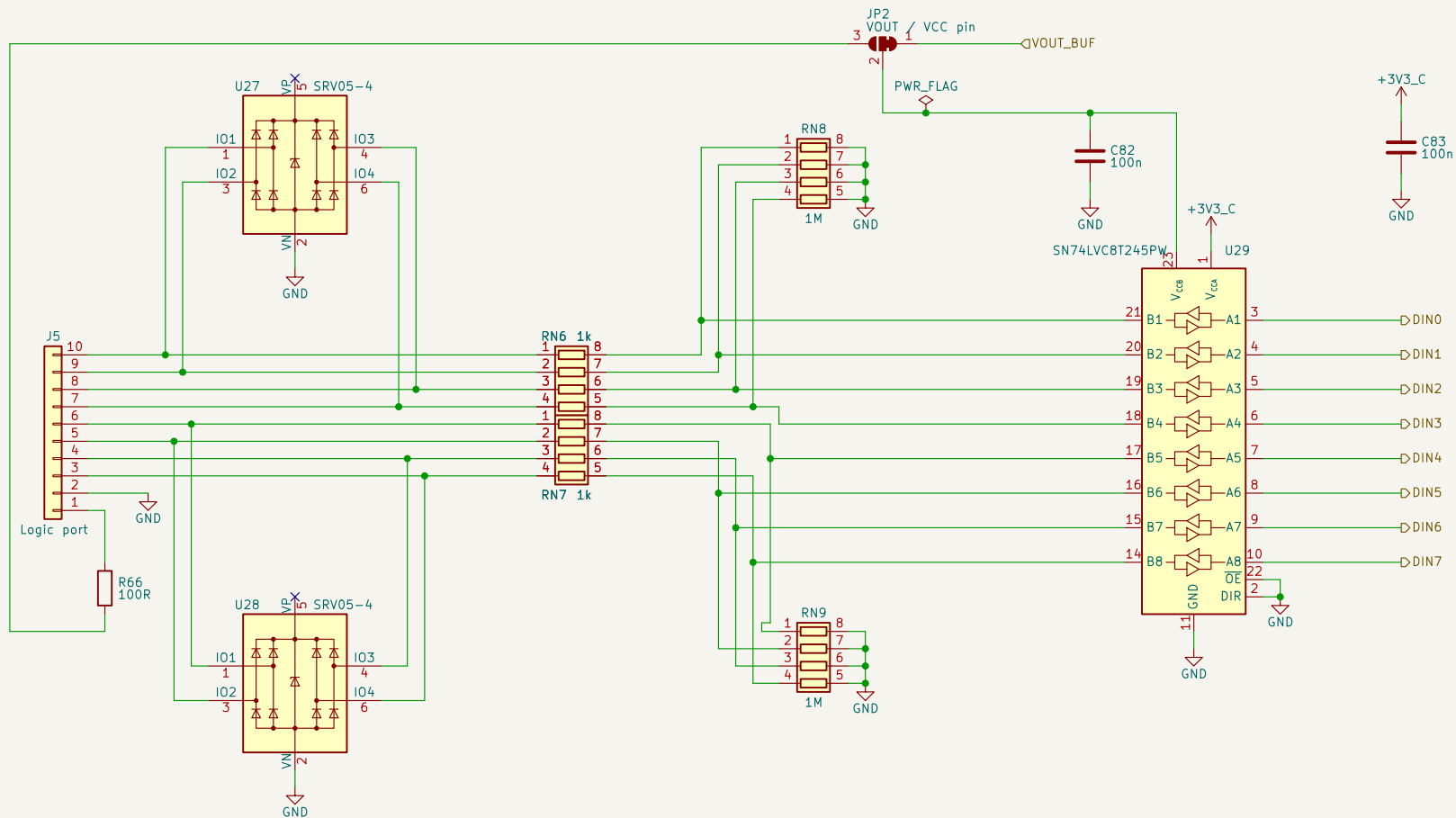


Step up at 90 mV, over-current at 120 mV, jump to range 3 at 150 mV across the shunt

Sheet: /Comparators/		
File: comparators.kicad_sch		
Title: Range and Over-current Comparators		
Size: A4	Date: 2026-10-08	Rev: A1 (draft)
KiCad E.D.A. 10.0.0		Id: 10/13



Sixteen bits captured with every sample: status first, then the eight logic inputs		
Sheet: /Side Data/ File: side_data.kicad_sch		
Title: Side Data: Status and Logic Inputs of Every Sample		
Size: A4	Date: 2026-10-08	Rev: A1 (draft)
KiCad E.D.A. 10.0.0	Id: 11/13	



J5 has the pin order of the PPK2 logic port: pin 1 VCC, pin 2 ground, pins 3 to 10 are logic inputs 7 down to 0.
 U29 port B is powered through JP2: from VOUT_BUF as built, so the inputs follow the DUT supply,
 or from the VCC pin of the port for a DUT whose logic runs on another voltage (1.65 V to 5.5 V).
 DIR and OE are referenced to port A (+3V3_C): DIR low sets the direction B to A.
 With port B unpowered the outputs float; pull-downs on the Side Data sheet hold DINO to DIN7.
 Each input loads the DUT pin with 1 M to ground.

Logic port in the pin order of the PPK2; the inputs follow the DUT logic levels

Sheet: /Digital Inputs/
 File: digital_inputs.kicad_sch

Title: Logic Port: Digital Inputs D0 to D7

Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.0

Rev: A1 (draft)

Id: 12/13

